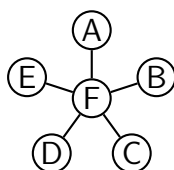


Weekly Quiz 3

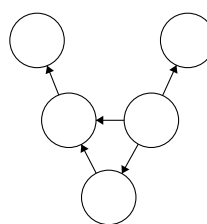
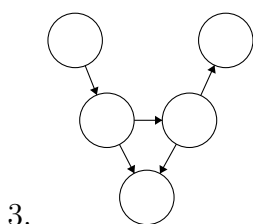
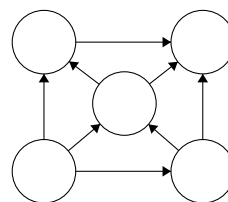
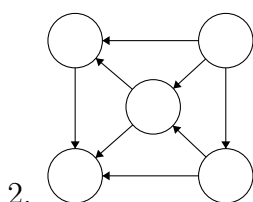
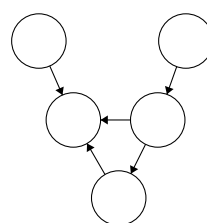
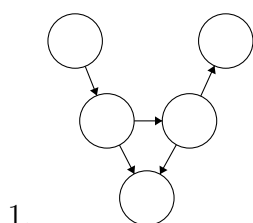
1. Alice has 6 distributions A, B, C, D, E, F. She decides to represent them using a Bayesian network, with edges from A, B, C, D, E to F as shown in the Figure below (she is currently unaware of the directions she has to set for these edges). What she does know is that there is no set of 2 variables that are unconditionally independent of each other. Out of the 32 different ways Alice can assign directions to the edges between A, B, C, D, E and F, how many of them do not violate this assumption?

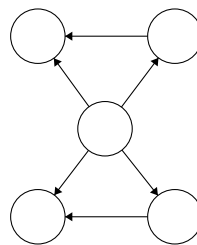
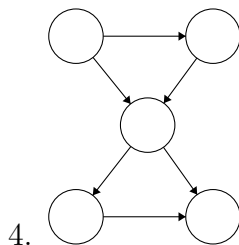


Solution:

There are 6 such ways. Since the variables are not unconditionally independent, F cannot be the center of a v-structure. Thus, no two of {A, B, C, D, E} can simultaneously be parents of F. This leaves 5 ways of choosing a single parent, or 1 way of having F be the parent of all others.

2. Which of the following pairs of BNs are equivalent?



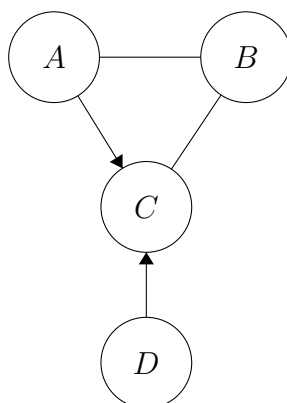


Solution:

1 and 2 do not contain equivalent BNs, 3 and 4 do. In 1, notice that a new immorality is introduced between the top left, the middle left and middle right nodes. In 2, an immorality is lost and a new one is created. In 3 and 4, there were no immoralities before or after, so the skeletons being the same guarantees the BNs are equivalent.

Instead of implicitly labelling the nodes, one valid interpretation was that the question asked if for some labelling of the nodes, the two BNs were equivalent. Under that interpretation, 2 would also contain equivalent BNs, since the structures are simply mirror images of each other. Both interpretations have been given marks.

3.
 1. The first part of the PC algorithm can build a skeleton (set of edges ignoring orientation) which can have more edges than if we construct a correct and minimal BN using the subset construction algorithm, for some ordering of variables. (assume the first part of PC algorithm can be run regardless of whether the distribution has a BN)
 2. Bob has run the PC algorithm on a distribution, and it returned a skeleton, as well as some orientations determined as per the algorithm. Bob believes he can reconstruct the true Bayesian Network **uniquely** from the information the PC algorithm has given. Does he have all the information he needs (assuming at least one perfect BN exists for the distribution)?

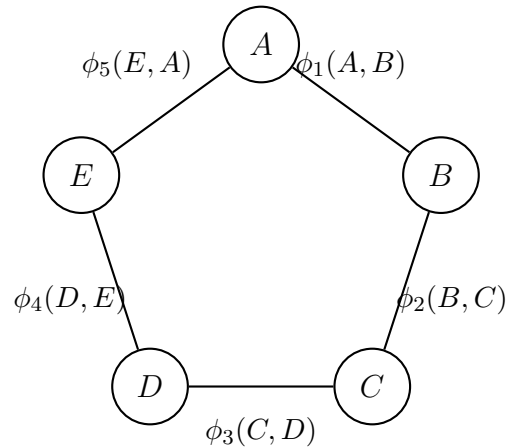


Solution:

1. False. If the PC algorithm determines that an edge does exist between A and B , it is because there is no subset of variables W such that $A \perp\!\!\!\perp B|W$. Then, if we assume W.L.O.G that B is added after A in the ordering, and F.T.S.O.C assume that in the correct and minimal BN, no edge exists between A and B , then there exists some set U with $A \notin U$ such that $B \perp\!\!\!\perp S \setminus U|U$, where S is the set of vertices added before B . This means that $B \perp\!\!\!\perp A|U$, which would make U a valid witness set. Thus, any edge in the PC skeleton is present in any minimal and correct BN for the distribution.
2. True. Note that if the B-C edge were oriented from B to C, a new immorality would be formed involving B, C and D. Since the PC algorithm is supposed to identify all immoralities in the equivalence class of the distribution, the edge must be oriented from C to B. Additionally, since A-C-B cannot form a cycle by the rules for valid BNs,

the A-B edge must be oriented from A-B. Thus, it is guaranteed that only one graph exists in the equivalence class of the perfect BN for this distribution.

4. Consider a version of the Student Misconception example discussed in class with 5 students, satisfying the symmetric dependencies $(A \perp\!\!\!\perp C, D|B, E)$, $(B \perp\!\!\!\perp D, E|A, C)$, $(C \perp\!\!\!\perp A, E|B, D)$, $(D \perp\!\!\!\perp B, A|C, E)$, $(E \perp\!\!\!\perp B, C|A, D)$. All variables are binary ($\{0, 1\}$)



Notice that we can assume the above clique potentials are the only ones to be considered for this question (no single-size cliques).

1. To represent this distribution without the Bayesian Network or Markov Network being present, how many **free** variables would be needed?
2. To represent this distribution with a minimal and correct **Bayesian Network**, how many **free** variables would be needed? Assume the BN is constructed with an ordering of A, B, C, D, E.
3. To represent this distribution with the above Markov Network (note that no assumptions are made that the clique factors represent probability distributions, and the above 5 clique potentials are the only ones to consider), how many **free** variables would be needed?

Solution:

1. $2^5 - 1 = 31$.
 2. Following the MRF, one can factorize the BN as $P(A)P(B|A)P(C|A, B)P(D|A, C)P(E|A, D)$, with $1+2+4+4+4=15$ conditional independencies. Earlier, 19 was given as the correct answer according to the wrong factorization $P(A)P(B|A)P(C|A, B)P(D|A, B, C)P(E|A, D)$, but since $B \perp\!\!\!\perp D|A, C$, it suffices to take $P(D|A, C)$.
 3. Note that it has been mentioned that the potentials need not be normalized, thus all 4 values for each potential are free variables. Parameters such as Z could be inferred from these values, and hence would not be free. The answer is $4*5 = 20$
5. Consider a probability distribution over variables $x_1, x_2, x_3, x_4, x_5, x_6, x_7$. The joint distribution can be specified as $P(x_1, x_2, x_3, x_4, x_5, x_6, x_7) \propto e^{x_1 \oplus x_2 \oplus x_3} e^{x_2 \oplus x_3 \oplus x_4 \oplus x_5} e^{x_2 \oplus x_6} e^{x_4 \oplus x_7}$. Which of the following CIs is/are true?
1. $x_1 \perp\!\!\!\perp x_6 | x_2, x_5$
 2. $x_2 \perp\!\!\!\perp x_5 | x_1, x_3, x_4$

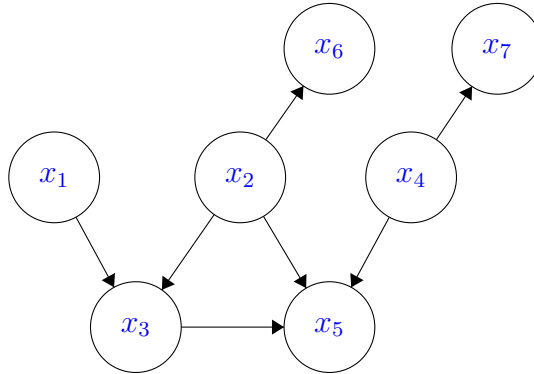
$$3. x_2 \perp\!\!\!\perp x_7 | x_3, x_4$$

$$4. x_1 \perp\!\!\!\perp x_5 | x_3, x_4$$

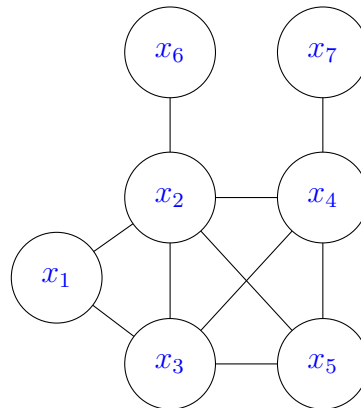
Solution:

(a), (c) are True. (b), (d) are False. Constructing either a BN or a MRF isn't automatically enough to rule out (b) or (d), but can be useful to prove (a) and (c) hold.

A BN for the distribution is given below:



A MRF for the distribution is given below. In (b), notice the path directly connecting x_2 and x_5 . In (d), notice the path via x_2 .

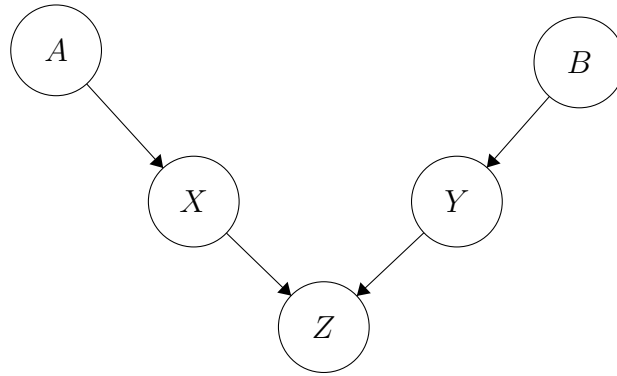


Alternatively, for (a), conditioned on the known variables, we can reinterpret the probability distribution as being proportional to $(F_1(x_1, x_3)F_2(x_3, x_4)F_4(x_4, x_7))(F_3(x_6))$. This is a product of functions of two disjoint sets, hence those two sets are independent. For (c), similarly, we have $(F_1(x_1, x_2)F_2(x_2, x_5)F_3(x_2, x_6))(F_4(x_7))$.

For (b), notice that one combination of x_2 and x_5 , which makes F_1 and F_2 both evaluate to e, will be e^2 times more probable than the assignment which makes them both evaluate to 0.

For (d), notice that, WLOG, the term $e^{x_1 \oplus x_2} e^{x_2 \oplus x_5}$ is what is of interest to us. The assignments which set $x_1 = x_5$ are $\frac{e^2+1}{2e}$ times more likely than the ones setting both to be different. Thus, using BNs, MRFs or the fact that $P \models (X \perp\!\!\!\perp Y | Z) \leftrightarrow P(X, Y, Z) = \phi_1(X, Z)\phi_2(Y, Z)$, we can prove which of the above 4 CIs are true. Proving which are not true is a bit trickier, but this particular distribution does not throw up any surprises, and the proofs are tractable.

6. Answer the following questions based on a run of the PC algorithm on a probability distribution that factorizes according to the below BN, and has no additional independencies not encoded by the BN.



1. How many Witness Sets are possible for X, Y which do not contain the node Z?
2. How many Witness Sets are possible for X, Y which contain the node Z?
3. How many Witness Sets are possible for A, B?

Solution:

1. There are 4 possible witness sets for X, Y. None of these can contain Z, since $X \not\perp\!\!\!\perp Y|Z$. Thus, 4 is the answer here.
2. As noted in the above subpart, all 4 possible witness sets do not contain Z. Thus, the answer here is 0.
3. The only configuration of X, Y, Z under which A and B are dependent is if Z alone is observed and the other two are unobserved. In the other 7 configurations, A and B are independent. Thus, there are 7 possible witness sets.